Alessio Sancetta

Curriculum Vitae

asancetta@gmail.com

Research Interests

Econometrics, finance, macroeconomics, causal inference, large datasets analysis, machine learning.

Leadership

Extensive experience managing teams and coaching individuals with more than two decades professional experience in the financial industry and higher education.

Academic Appointments

2012.07–Present **Chair of Economics**, Department of Economics, Royal Holloway, University of London, UK
Tenured Professor of Economics, Director of the MSc in Finance and Co-Director of the MSc in Computational Finance (jointly with Computer Science) (2012.09-2021.08), Co-Director and Co-Founder of the Centre for Robust Inference in a Digital Economy (RIDE).

2017.09–2020.09 **Adjunct Professor**, Department of Business and Management, LUISS, Rome, IT
Course leader for the module Quantitative Methods for Finance in the MSc Corporate Finance (laurea magistrale).

2005.10–2008.06 **University Lecturer**, Department of Economics, University of Cambridge, UK

2002.11–2005.09 **Research Associate**, Department of Applied Economics, University of Cambridge, UK

Education

2000–2002 **PhD**, *Trinity College, University of Cambridge*, UK
Thesis: *Approximation Methods in Financial Econometrics.*

1999–2000 **MSc**, *University of York*, UK
Rank 1 out of 120 MSc students in the Economics Department.

1995-1997 **BSc**, *Fairfield University*, USA
Rank 1 out of the whole university BS graduating class.

Industry Experience

2021.08–2022.04 **Quantitative Research & Development Lead, Strategy and Planning Department**, ADIA, Abu Dhabi
In charge of developing a systematic review process for all low frequency, high capacity quantitative strategies within a newly established quant group, resulting in a decision on whether a strategy is viable for a cash allocation.

2017.10–2021.06 **Head of Algorithmic Execution Intelligence, FRC**, UBS, London
Headed a team in charge of developing models, strategies, pre-trade transaction cost analysis, and high frequency alpha signals using financial econometrics and machine learning for execution algorithms in foreign exchange and futures.

2011.04–2014.03 **Quantitative Trader**, London
Partner and consultant in various algorithmic trading projects. Inter alia, projects included building a client flow business and consulting for a Chicago based team (ex TransMarket Group) on high frequency alpha generation for foreign exchange, futures on CME (foreign exchange, energy, and equity indices) and US equities and ETF's.

2009.11–2011.03 **Head of Proprietary Trading**, Celeritas Markets, London
Devised high frequency alpha signals and strategies in foreign exchange with daily volumes in the top 10 worldwide on Reuters and top 20 worldwide on EBS with live Sharpe ratio in excess of 8.

2009.05–2009.11 **Algo Trader**, Deutsche Bank, London
Managed risk of existing high frequency proprietary algorithms in foreign exchange, conducted research to automate the calibration process of the algorithms, devised alpha signals.

2008.05–2009.02 **Financial Econometrics Specialist**, BNP Paribas, London
Conducted research on volatility estimation and prediction, worked on volatility trading strategies and arbitrage, and backtesting environment for vanilla foreign exchange options strategies.

2004.11–2008.06 **Independent Portfolio Manager**, Part Time
Managed a small hedge fund that traded in liquid futures and options. Devised intraday and medium frequency strategies based on volatility, price reversals and options mispricing.

2003.09–2006.12 **Research Advisor**, Dharma Holdings Luxembourg, Part Time
In charge of the modelling and mathematical development of a computerized risk management system for stocks and derivatives.

1998.05–1999.07 **Trader of Derivatives**, European Investment Management, Dharma Holdings Group, Rome
Intra-daily semi-discretionary trading of listed futures and vanilla options based on proprietary indicators and models.

Languages

Italian (native), English (fluent), Spanish (basic), French (basic reading).

Computer Skills

Matlab, Python, R, Java (basic). OS: Linux, Mac.

Listing and Awards

2016 Econometric Theory Multa Scripsit Award.

2007 Biography in Marquis Who's Who in the World 2007.

2000 Prize for the MSc with the highest final mark in the Economics Department, University of York.

1997 Prize for the Bachelor of Science with the highest GPA in the whole university, Fairfield University.

Press Coverage

The following co-authored papers have been cited in the press:
"Price Drift before U.S. Macroeconomic News: Private Information about Public Announcements?" (inter alia FT, CNN, Bloomberg, Wall Street Journal);
"Drift Begone! Release Policies and Preannouncement Informed Trading" (Telegraph, Wall Street Journal).

Publications

Internationally Refereed

[1] Molten Lava Meets Market Languor (with Steve Satchell), **Quantitative Finance** 2 (2002), (comments section) 405.

[2] The Bernstein Copula Function as a Model and Approximation to Multivariate Distributions (with Steve Satchell). **Econometric Theory** 20 (2004), 535-562.

[3] Calculating Hedge Fund Risk: The Draw Down and the Maximum Draw Down (with Steve Satchell). **Applied Mathematical Finance** 11 (2004), 259-282.

[4] New Test Statistics for Market Timing with Applications to Emerging Markets (with Steve Satchell). **The European Journal of Finance** 11 (2005), 419-443.

[5] Distance between Non-Identically Weakly Dependent Random Vectors and Gaussian Random Vectors under the Bounded Lipschitz Metric. **Statistics and Probability Letters** 75 (2005), 158-168.

[6] Changing Correlation and Equity Portfolio Diversification Failure for Linear Factor Models during Market Declines (with Steve Satchell). **Applied Mathematical Finance** 14 (2007), 227-242.

[7] Weak Convergence of Laws on RK with Common Marginals. **Journal of Theoretical Probability** 20 (2007), 371-380.

[8] Online Forecast Combinations of Distributions: Worst Case Bounds. **Journal of Econometrics** 141 (2007), 621-651.

[9] Nonparametric Estimation of Distributions with Given Marginals via Bernstein-Kantorovich Polynomials: L1 and Pointwise Convergence Theory. **Journal of Multivariate Analysis** 98 (2007), 1376-1390.

[10] Sample Covariance Shrinkage for High Dimensional Dependent Data. **Journal of Multivariate Analysis** 99 (2008), 949-967.

[11] Strong Law of Large Numbers for Pairwise Positive Quadrant Dependent Random Variables. **Statistical Inference for Stochastic Processes** 12 (2009), 55-64.

[12] Maximal Inequalities for U-Processes of Strongly Mixing Random Variables. **Probability and Mathematical Statistics** 29 (2009), 155-167.

[13] Consistent Estimation of a General Nonparametric Regression Function in Time Series (with O.B. Linton). **Journal of Econometrics** (Peter Robinson's Festschrift) 152 (2009), 70-78.

[14] Forecasting and Prequential Validation for Time Varying Meta-Elliptical Distributions (with A. Nikandrova). **Journal of Time Series Econometrics** 1 (2009), Iss. 2, Article 1.

[15] Nearest Neighbor Conditional Estimation for Harris Recurrent Markov Chains. **Journal of Multivariate Analysis** 100 (2009), 2224-2236.

[16] Bootstrap Model Selection for Possibly Dependent and Heterogeneous Data. **Annals of the Institute of Statistical Mathematics** 62 (2010), 515-546.

[17] Recursive Forecast Combination for Dependent Heterogeneous Data. **Econometric Theory** 26 (2010), 598-631.

[18] Universality of Bayesian Predictions. **Bayesian Analysis** 7 (2012), Number 1, 1-36, 45-46 (rejoinder).

[19] Weak Conditions for Nonparametric Density Shrinkage. **Journal of Multivariate Analysis** 115 (2013), 285-300

[20] Conditional Estimation for Dependent Functional Data (with Heather Battey). **Journal of Multivariate Analysis** 120 (2013), 1-17.

[21] A Recursive Algorithm for Mixture of Densities Estimation. **IEEE Transactions on Information Theory** 59 (2013), 6893-6906.

[22] An open Problem on Strongly Consistent Learning of the Best Prediction for Gaussian Processes (with Laci Györfi). **Topics in Nonparametric Statistics** Proceedings of the First Conference of the International Society for Nonparametric Statistics (2014).

[23] Semiparametric Estimation of a Class of Generalized Linear Models without Smoothing. **Journal of Multivariate Analysis** 130 (2014), 141-154.

[24] A Nonparametric Estimator of the Covariance Function for Functional Data. **Econometric Theory** 31 (2015), 1359-1381.

[25] Greedy Algorithms for Predictions. **Bernoulli** 22 (2016), 1227-1277.

[26] Estimation for the Prediction of Point Processes with Many Covariates. **Econometric Theory** 34 (2018), 598-627.

[27] Inference for Constrained Stationary Autoregressive Processes. **IEEE Transactions on Information Theory** 65 (2019), 538-550.

[28] Price Drift before U.S. Macroeconomic News: Private Information about Public Announcements? (with Alexander Kurov, Georg Strasser and Marketa Wolfe). **Journal of Financial and Quantitative Analysis** 54 (2019), 449-479.

[29] Intra-Day Prediction of End of Day Trading Volume. **Journal of Financial Econometrics** 19 (2021), 472-495.

[30] Estimation in Reproducing Kernel Hilbert Spaces with Possibly Dependent Data. **IEEE Transactions on Information Theory** 67 (2021), 1782-1795.

[31] Estimation of a High Dimensional Counting Process Without Penalty for High Frequency Events (with Luca Mucciante). **Econometric Theory** (2022) online. `https://doi.org/10.1017/S0266466622000238`.

[32] Drift Begone! Release Policies and Preannouncement Informed Trading (with Alexander Kurov, and Marketa Wolfe). **Journal of International Money and Finance** 128 (2022) online first. `https://doi.org/10.1016/j.jimonfin.2022.102718`.

[33] Testing Subspace Restrictions in the Presence of High Dimensional Nuisance Parameters. **Electronic Journal of Statistics** 16 (2022), 3751-3788..

[34] Intraday Trades Profile Estimation: An Intensity Approach. **Journal of Financial Econometrics** 21 (2023), 651-677.

[35] Empirical Asset Pricing with Functional Factors (with Philip Nadler). **Journal of Financial Econometrics** 21 (2023), 1258-1281.

[36] An Order Book Dependent Hawkes Process for Large Datasets (with Luca Mucciante). **Journal of Financial Econometrics** (2023) online first. `https://doi.org/10.1093/jjfinec/nbad021`

[37] Consistent Causal Inference for High-Dimensional Time Series (with Francesco Cordoni). **Journal of Econometrics** 246 (2024). `https://doi.org/10.1016/j.jeconom.2024.105902`

Book Chapter

Draw Downs as a Measure of Hedge Fund Risk: Some Stylized Facts. (with Gianfranco Lande and Steve Satchell). In G.N. Gregoriou, F. Rouah, and V.N. Karavas (editors), **Hedge Funds: Strategies, Risk Assessment, and Returns**. Washington: Beard Books, 2003, 235-246.

Work in Progress

[1] Action-State Dependent Dynamic Model Selection (with Francesco Cordoni).`https://arxiv.org/abs/2307.04754`

[2] Information Content of Book and Trade Order Flow at Different Trading Volume Time Scales (with Mariol Jonuzaj and Yuri Taranenko).https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4971656

[3] High Dimensional Causal Projection Estimators.

[4] What Threshold Should be Applied to Statistical Tests in Finance? (with Cam Harvey and Yuqian Zhao).

[5] A Simple Approach to Asset Pricing with Weak Factors (with Yuqian Zhao).

[6] Assessing the Impact of Book and Trade Order Flow with a Hawkes Processes in a Random Environment.https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5170318

Working Papers

[1] Bernstein Approximations to the Copula Function and Portfolio Optimization (with Steve Satchell), Cambridge Working Papers in Economics (2001), 0105.

[2] Cost of Capital and the Regulator's Preferences: An Investigation into a New Method of Estimating Regulatory Beta (with Steve Satchell). Cambridge Working Papers in Economics (2004), 0441.

[3] Copula Based Monte Carlo Integration in Financial Problems. Cambridge Working Papers in Economics (2005), 0506.

Editorial Work

2007.05–2008.08 **Assistant Editor and Book Review Editor**, The Econometrics Journal

Major Grants

2022.10–2025.09 **Information Content and Dissemination in High-Frequency Trading**, Leverhulme Trust RPG-2021-359, (PI with Francesco Feri and Michael Naef as CoI's)

2003.10–2005.09 **Econometric Modelling of Energy and Soft Commodity Prices with Applications**, ESRC RES-000-23-0400, (with Steve Satchell)

Refereeing

Annals of Statistics, Australian and New Zealand Journal of Statistics, Biometrika, Econometric Theory, IEEE Transactions on Information Theory, JASA, Journal

of Applied Econometrics, Journal of Business and Economic Statistics, Journal of Financial Econometrics, Journal of Financial Forecasting, Journal of Multivariate Analysis, Journal of Nonparametric Statistics, Journal of Time Series Analysis, Review of Economics and Statistics, Statistics and Probability Letters, TEST.

Additional Interests

Creative writing, philosophy, literature, caving, scuba diving.